

WHEN CONSENSUS IS NOT ENOUGH: SELECTION, FEEDBACK, AND RECOVERY IN A FROZEN WORLD-ACTION MODEL

Anonymous authors

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ABSTRACT

World-action models jointly predict actions, future observations, and values, suggesting that test-time uncertainty or sample consensus could improve closed-loop control. We investigate when these signals support a useful decision, using a frozen Cosmos Policy on explicitly defined LIBERO-PRO subsets. A six-controller, 199-configuration comparison finds no statistically convincing improvement over joint max-value selection. A decoder-space medoid study with stronger initial-pool controls matches max-value at 112/180 successes; shorter execution horizons do not improve average success. Frozen-candidate experiments separate selection opportunity from stochastic continuation: one third of 288 saved candidates change outcome across suffix seeds, although a narrowly replicated pool exhibits persistent value misranking. Feedback and RGB-based regrasp improve success in selected settings, including a 27.5 percentage-point recovery gain on new initializations of eight difficult cells, without established transfer to new cells. Finally, paired observation-contract experiments identify 16 lost successes after an extra regrasp releases an already coupled object. These results distinguish state-risk detection, within-pool ranking, and intervention advantage. They support a controlled diagnostic account of inference-time planning, not a universally superior selector or a new benchmark state of the art.

1 INTRODUCTION

World-action models predict robot actions together with their possible consequences (Kim et al., 2026). This suggests selecting the most valuable candidate, penalizing uncertainty, or following a representative mode of the generated distribution. Yet agreement can reflect a shared error, and predicting a consequence accurately does not make that consequence desirable. A score distinguishing easy from difficult scenes may not identify the best action within one scene.

In out-of-distribution manipulation, failures can arise from poor ranking, a candidate pool with no useful action, insufficient feedback, or harmful recovery. Querying more often changes both information and action commitment. Gathering diagnostic evidence through a physical probe can itself alter contact. Improving one metric cannot isolate these mechanisms.

We investigate them using one frozen Cosmos Policy checkpoint and a series of deployed-controller and saved-state experiments. Our empirical contributions are (i) scoped comparisons of geometric and decoder consensus with joint max-value, including fixed-index and horizon controls; (ii) repeated-candidate diagnostics of opportunity and continuation variability; and (iii) paired analysis of feedback and RGB recovery, including intervention-induced harm. We retain unsuccessful methods and replay limitations rather than pooling heterogeneous runs into one success rate. The goal is a diagnostic account, not a claim that consensus or uncertainty methods are generally ineffective.

2 RELATED WORK

World models and selection. Diffusion Policy generates actions with receding-horizon execution (Chi et al., 2023), while Diffuser generates trajectories for planning (Janner et al., 2022). TD-MPC2

Table 1: Different evaluation populations. Reused controls, shared prefixes, and suffix repeats are not additional independent scenes.

Study	Population per controller	K	T/H
Broad selectors	199: 50 Object, 50 Environment, 99 Position	4	16/16
Decoder main	180: 60 per factor	3	16/16
Decoder H8	120 matched: 30/30/60 by factor	3	16/8
Candidate replay	36 snapshots, 10 task/init clusters	8	16/16
RGB recovery	40 new-init branches, eight selected cells	4	16/8
Observation contract	96 prefixes, two suffixes, 48 init clusters	4	16/8

and Dreamer study learned world models for control (Hansen et al., 2024; Hafner et al., 2025). Here the backbone is frozen, and most inference jointly generates actions, futures, and values. This differs from the full autoregressive, rollout-finetuned planning system in the Cosmos paper.

Self-consistency aggregates language-model samples (Wang et al., 2023). KeyStone uses geometric clustering and medoid selection, whereas KDPE uses a density-based endpoint criterion (Dai et al., 2026; Rosasco et al., 2025). Consensus itself is not our novelty. Our small-population adaptations to Cosmos differ from their backbones, action parameterizations, and budgets.

Failure monitoring and intervention. Flow-based VLA uncertainty has been studied with independently trained velocity-field models and active fine-tuning (Römer et al., 2026), not merely noise samples from one checkpoint. Sentinel separates temporal inconsistency from lack of task progress (Agia et al., 2025); Rewind-IL combines overlap monitoring with recovery checkpoints (Zheng et al., 2026). Thus, neither overlap alarms nor detection followed by recovery are new ideas by themselves. UNISafe and StressDream investigate safety filtering and failure-oriented world-model use in distinct learning settings (Seo et al., 2025; 2026). Visual alignment and memory offer other robustness mechanisms (Kachaev et al., 2025; Cherepanov et al., 2026). Our study asks what the tested frozen model supports before attributing gains to an uncertainty formula.

3 PROBLEM AND EVALUATION CONTRACTS

Model and control. At physical time t_q , query q receives real external/wrist RGB, robot proprioception, and a T5 instruction embedding, denoted x_q . Full simulator state s_q is used only for replay and offline diagnostics. Joint inference samples

$$(A_q^{(i)}, \hat{I}_{q,T}^{(i)}, \hat{p}_{q,T}^{(i)}, V_q^{(i)}) \sim p_\theta(\cdot | x_q; \xi_i), \quad A_q^{(i)} \in \mathbb{R}^{T \times 7}, \quad i = 1, \dots, K. \quad (1)$$

The controller executes the first $H \leq T = 16$ actions of one candidate, then obtains new real observations. Generated images do not replace actual camera input at the next ordinary query. Native actions contain translation, rotation-vector, and gripper commands. Nine-dimensional proprioception is robot state, not object poses. Most experiments use joint-value best-of- K , not the complete original autoregressive $a \rightarrow s' \rightarrow v$ planning protocol.

Benchmark and populations. We evaluate the LIBERO-PRO Object task suite under Object, Environment, and Position perturbations (Liu et al., 2023; Zhou et al., 2025). A task suite and a perturbation factor are different identifiers. Table 1 separates the main populations. Full-controller studies use five denoising steps and 280 physical actions; settling is separate. Recovery consumes the remaining action budget. These are explicit subsets and intervention populations, not the full official benchmark.

Outcomes and inference. Success Y is the environment’s task predicate within budget. Failure does not automatically mean a drop: timeout and incomplete placement differ. Our local drop proxy

is not an official LIBERO-Safety score.

$$\text{MacroSR}_m = \frac{1}{3} \sum_f \frac{\sum_e Y_{m,f,e}}{n_f}, \quad \text{MicroSR}_m = \frac{\sum_{f,e} Y_{m,f,e}}{\sum_f n_f}. \quad (2)$$

We report paired effects, rescues (control fails, method succeeds), and harms (the converse). Intervals preserve study-specific task/init clusters and factor weights (Appendix E). Unadjusted intervals and corrected tests are distinguished; repeated seeds are not independent tasks (Agarwal et al., 2021). Post-hoc findings are not confirmatory tests.

Replay contracts. Initial simulator states match in the 199-configuration study, but initial $K = 4$ pools are bitwise identical in only approximately 31–46% of comparisons. That study compares deployed controllers, not selectors on perfectly fixed pools. The later decoder study uses cached initial RGB, proprioception, and candidates: all 180 comparison groups pass exact query-zero checks. After a different action, fresh observations legitimately diverge. Saved-state studies instead control snapshots, executed candidates, and suffix seeds explicitly.

4 SIGNALS AND CONTROLLERS

Disagreement. Across samples of one query, we compute

$$\sigma_{h,d} = \sqrt{\frac{1}{K} \sum_i (a_{h,d}^{(i)} - \bar{a}_{h,d})^2}, \quad U_{\text{first}} = \sqrt{\sum_{d=1}^6 \sigma_{1,d}^2}, \quad U_V = \text{Std}_i(V_i). \quad (3)$$

The first-action statistic excludes the gripper. Internal disagreement among repeated entries of one latent frame is a separate diagnostic, not an independent ensemble. Appendix A specifies aggregation axes and online availability. A candidate-specific penalty is $S_i = V_i - \lambda U_i$. Subtracting the same pool-wide uncertainty from every candidate cannot change the argmax; pool risk can instead inform a control-mode decision.

Geometric consensus. OSC-medoid integrates clipped translation commands and composes scaled rotations into trajectory proxies, then combines translation, rotation geodesic, and gripper-sign differences with temporal discount. These are command proxies, not contact-aware future poses. For candidate distance D ,

$$i_D = \arg \min_i \frac{1}{K-1} \sum_{j \neq i} D(i,j), \quad i_V = \arg \max_i V_i. \quad (4)$$

We execute a sampled chunk, not an average of incompatible modes. Controls include raw medoid, KeyStone-style guarded clustering, and a KDPE-inspired kernel on induced endpoints. Appendix B gives parameters.

Decoder medoid. We adapt an external decoder-token selection implementation (Double05, 2026) to Cosmos’s tiled action latent. Hidden features come from the last conditional denoising call before the output layer. The 196 spatial patches are not action-time tokens. If $C_{p,h}$ counts decoded action scalars associated with patch p and action time h , use

$$w_p = \frac{\sum_h C_{p,h} \omega_h}{\sum_{p',h} C_{p',h} \omega_h}, \quad D_Z(i,j) = \sum_p w_p [1 - \cos(z_{i,p}, z_{j,p})]. \quad (5)$$

Full weights give fourfold weight to actions 1–4 and unit weight to 5–16. Prefix-H8 sets weights 9–16 to zero. Selection uses Eq. (4) with stable lowest-index ties. Fixed-index controls test seed preference, and action medoid controls for representation.

Feedback and recovery. We distinguish committing a saved suffix, querying on the old observation (stale), and querying on current observations (fresh). These controls separate extra inference from fresh information. Recovery adds a physical open–retreat–relocalize–approach–close–lift sequence, then resumes Cosmos. A frozen RGB localizer and confidence/workspace/reach-miss gate enable non-privileged execution. Probe variants move the end effector before deciding whether to recover: their information-gathering action also changes contact.

Table 2: Broad deployed comparison. Factor and macro columns are percentages; the last column is successes out of 199. K1 is the single-sample control.

Method	Object	Env.	Position	Macro	Success
K1	94.00	38.00	28.28	53.43	94
Max-value K4	94.00	40.00	30.30	54.77	97
OSC-medoid K4	96.00	40.00	26.26	54.09	94
Raw-medoid K4	94.00	42.00	28.28	54.76	96
KeyStone-style K4	98.00	40.00	29.29	55.76	98
KDPE endpoint K4	94.00	40.00	25.25	53.08	92

Table 3: Decoder H16 and fixed-index controls. Each factor has 60 cases; total is out of 180. Indices are zero-based.

Controller	Obj.	Env.	Pos.	Total	Macro (%)
First / K1	56	24	25	105	58.33
Max-value	57	24	31	112	62.22
Action medoid	57	24	29	110	61.11
Decoder medoid	57	26	29	112	62.22
Fixed index 1	57	25	28	110	61.11
Fixed index 2	56	24	29	109	60.56

5 RESULTS

5.1 BROAD SELECTION DOES NOT ESTABLISH A GAIN

Six controllers on 199 configurations yield 1, 194 rollouts; one empty upstream initialization asset is excluded uniformly. Table 2 shows factor and macro rates. OSC-medoid changes macro success by -0.68 percentage points (pp), interval $[-3.33, 1.99]$, 3 rescues/6 harms. KeyStone-style has the best point estimate, but $+1.00$ pp has interval $[-2.01, 4.33]$. Raw medoid (-0.01 pp) and KDPE endpoint (-1.68 pp) also do not establish superiority. This concerns the tested subset and adaptations, not the original methods under their published protocols.

A residual-risk study illustrates the distinction between detection and ranking. Across 200 states/800 candidate branches, global failure AUROC is 0.650 but within-pool ranking is 0.509. The score reduces measured future-proprio error by 2.14%, yet success changes 117/200 to 113/200. Six replay-integrity deviations limit strict causal interpretation of this historical study. Separately, a frozen-ranker test improves its offline objective but gives 165/360 versus 164/360 closed-loop successes. State-difficulty prediction does not ensure action-advantage prediction.

5.2 DECODER CONSENSUS, SEEDS, AND EXECUTION HORIZON

The decoder campaign contains 720 main, 360 fixed-index, and 360 H8 rollouts. Decoder medoid and max-value both succeed 112/180 times (62.22%), with 7 rescues/7 harms and paired macro interval $[-3.33, 3.33]$ pp. Two additional Environment successes are offset by Position losses. Fixed-index controls achieve 110/180 and 109/180; differences from decoder also have intervals including zero (Table 3).

The matched H8 subset has 120 configurations with 30/30/60 factor counts. Equal-factor macro success is 65.00% for max-value H16, 59.44% for max H8, 57.78% for full decoder H8, and 59.44% for prefix decoder H8. Principal paired intervals include zero (Appendix C), and candidate calls roughly double. Neither shortening execution nor prefix weighting establishes a benefit; this does not prove universal harm from H8.

Within each of three fixed candidate seed groups, one index accounts for 81–84% of decoder queries (Fig. 1, left). Forty of 180 episodes use one index throughout, although most are not constant. This motivates new seed-bank and permutation controls before treating hidden similarity as physical correctness. It does not prove seed identity caused every observed outcome.

Table 4: Observation-contract outcomes. Drops are a local proxy, not official LIBERO-Safety. These branches share 96 prefixes and 48 initclusters.

Controller	Success / 192	SR (%)	Drops	Regrasps
Continue H8	128	66.67	5	0
Physical regrasp	151	78.65	7	70
Open refresh only	154	80.21	7	70
Preserve refresh only	161	83.85	7	70
Open refresh + regrasp	138	71.88	23	90
Preserve refresh + regrasp	145	75.52	23	90
Oracle: calibrated	156	81.25	1	104
Oracle: physical	167	86.98	6	134

5.3 OPPORTUNITY IS CONDITIONAL ON THE CONTINUATION

Eight saved candidates from each of 36 snapshots initially give 18/36 max-value successes and 19/36 empirical-oracle successes; 17 pools contain no successful tested branch. The empirical oracle is

$$O_K(s) = \max_i Y(s, A_i, \xi_{suffix}). \quad (6)$$

It is conditional on its suffix, not a universal upper bound on attainable success.

Three additional continuation seeds change labels for 96 of 288 fixed candidates. Six initially all-fail pools then contain a success. Choosing and evaluating on the same repeats yields an optimistic 55.56%; held-out-suffix selection gives 40.74%, equal to max-value. Outcome-based selection also needs holdout.

Persistent misranking exists locally. In one saved pool, two alternatives each succeed 10/10 times on new suffixes while the max-value candidate succeeds 1/10 (Holm-adjusted $p = 0.01172$). These are two actions in one state, not independent tasks. On eight neighboring pools, held-out-repeat selection gives 40/80 versus 36/80, interval $[-3.75, 18.75]$ pp. It still uses offline outcome information; a transferable online verifier remains absent.

5.4 NARROW POSITIVE FEEDBACK AND RGB RECOVERY

A shared-prefix Object/task 0 study branches at $t = 72$, after eight actions of a common query at $t = 64$. Fresh requery versus committing the remaining eight actions gives 64/100 versus 46/100 successes: +18 pp, init-cluster interval $[4, 32]$. The 100 branches contain 50 initializations with two rollout seeds. This supports that intervention on that task, not a universal time 72 schedule.

A separate 36-snapshot repeat-feedback study gives 41/108 fresh 8 successes, 44/108 open 16, and 45/108 stale 8. Fresh-minus-stale is -3.70 pp, interval $[-20.51, 9.72]$. Matched- K /continuity tests on 24 states give 15 open 16, 14 stale, and 13 for each fresh variant. More frequent observation alone is not an established remedy.

RGB regrasp provides the strongest narrow improvement: on new initializations of eight selected difficult Position cells, continuation succeeds 13/40 times and RGB-triggered recovery 24/40, a +27.5 pp effect, init-cluster interval $[12.5, 42.5]$, 12 rescues/1 harm. Same-cell replication gives 6/40 versus 25/40. Transfer to seven other cells gives 26/35 versus 31/35, interval $[-2.9, 31.4]$ pp. These cells were not necessarily unseen during localizer training; general transfer is not established. A later router adds only two successes over full recovery (55/75 versus 53/75), so the full system's gain cannot be assigned to the router.

5.5 OBSERVATION CONTRACTS REVEAL INTERVENTION-INDUCED HARM

A development recovery experiment reuses 96 prefixes from 12 selected Position cells, with two suffixes and eight arms: 1536 branches, 48 initclusters. When the old gate declines recovery, a 3 action probe either opens the gripper or preserves its previous command, followed by refresh only or extra regrasp. Refresh-only arms still keep old-gate recovery on previously eligible states. Oracle controls use privileged localization/workspace information.

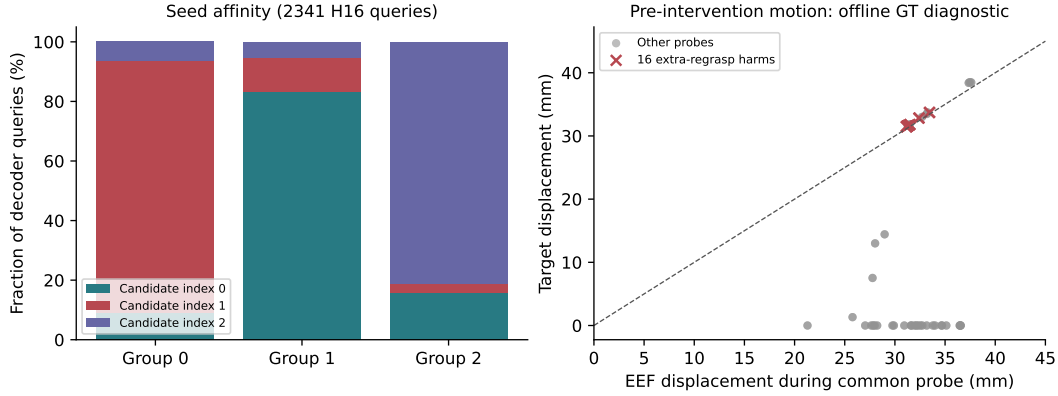


Figure 1: Two diagnostics from different populations. Left: candidate-index frequencies across 2341 decoder queries reveal seed-group affinity. Right: target and EEF displacement during the common probe; red marks the 16 branches later harmed by extra regrasp. Their objects already move with the gripper. Motion is an offline GT diagnostic. Repeated suffixes can overlap in this plot.

The primary preserve-refresh-plus-regrasp arm succeeds 145/192 times versus 151/192 for old physical recovery: -3.125 pp, interval $[-6.25, 0]$, 9 rescues/15 harms, Holm $p = 1$. None of six pre-declared pooled contrasts passes Holm 0.05. Post-hoc preserve-only achieves 161/192, but its ten net successes over physical recovery are concentrated in one cell; cluster-sign $p = 0.09587$ does not establish transfer.

Adding regrasp after the matched preserve probe loses 16 successes and rescues none, across eight prefixes and six initclusters. Before divergence, the target moves 32.01 mm and the EEF 31.68 mm on average, with contact throughout all three common probe steps. Regrasp opens the gripper near $t = 76$; drops occur at $t = 82-85$ during the maneuver. Refresh-only succeeds in all 16. The ground-truth motion diagnostic (Fig. 1, right) is offline, not an online RGB input. The paired effect is -8.33 pp, interval $[-11.46, -5.21]$, while cluster-sign $p = 0.03158$ becomes 0.18948 after Holm. This is a localized audited mechanism, not sixteen independent confirmations of a general safety claim.

5.6 FINAL RECOVERY CONFIRMATION AND GROUNDING DIAGNOSTIC

The completed confirmation contains 128 paired cases and four arms (512 branches), split equally between familiar cells and an x0.3 transfer cohort. At the historical branch time $t = 72$, physical RGB recovery succeeds 41/64 times on familiar cells, versus 16/64 for matched H8 continuation and 17/64 for H16 max-value continuation. The recovery-minus-H8 effect is $+39.06$ pp, with an init-cluster interval $[29.69, 48.44]$ and 27 rescues versus two harms. This confirms a local recovery effect, not a universal intervention schedule. Preserve-only also achieves 41/64 and does not improve the physical recovery baseline.

On x0.3 transfer, both continuation controls achieve 0/64, and physical recovery and preserve-only each achieve 1/64. A timing sweep at $t = 56, 72, 88$ reuses these same prefixes (768 branches): familiar-cell physical recovery achieves 31/64, 41/64, and 34/64, respectively. These timing comparisons are dependent and do not establish a universal optimum at $t = 72$.

A separate completed privileged diagnostic uses 16 cases in each cohort and four recovery variants (128 branches). On transfer, replacing predicted XY with ground-truth XY while keeping the fixed RGB gates raises SR from 0/16 to 3/16. Ground-truth XYZ gives 2/16, with or without physical gating. Geometry checks find negligible GT projection round-trip error, whereas transfer RGB localization is poor. Thus localization contributes to failure, but correcting it alone leaves 13/16 transfer failures. Oracle results are mechanistic diagnostics, not deployable-policy scores. A broad, paired valid199 comparison of historical and observation-triggered P3 is pending; no aggregate P3 benchmark score is claimed here.

6 DISCUSSION AND LIMITATIONS

Risk, choice, and intervention require distinct evidence. A detector can separate scene difficulty without ranking actions. Consensus can identify a likely but physically incorrect mode. Requery can interrupt a useful sequence, and regrasp can destroy a useful contact. Intervention-specific advantage should include the cost and harm of gathering information.

The study is limited to one frozen backbone in simulation. Selected cells, reused prefixes, and few independent clusters limit generalization. K , horizons, and perturbation levels differ across campaigns: cross-table SR differences are not method effects. The broad selector study has imperfect initial-pool matching. Narrow gains do not establish new-task or real-robot transfer; oracles use unavailable information. Shared-server timings are not controlled latency measurements. Repeated exploration requires new holdouts to prevent selection bias. These constraints do not establish impossibility for other uncertainty, consensus, or feedback methods.

7 CONCLUSION

Neither geometric nor decoder consensus establishes a broad improvement over joint max-value for the tested frozen Cosmos Policy. Candidate repeats reveal suffix-dependent labels and narrow persistent misranking. Feedback and RGB recovery help in selected states, but extra intervention can cause physical failure. The next priority is to predict intervention-specific advantage and harm on new cluster-separated data while auditing proposal coverage, ranking, information, and recovery separately.

REPRODUCIBILITY STATEMENT

The working artifact contains source-linked tables, generation scripts, checksums, manifests, and paired video galleries. Appendices specify inference, aggregation, and replay checks. The final audit decoded 3000 videos/480434 frames and checked 23933 decoder queries, including dependent branches and smoke tests. The manuscript source package is self-contained for typesetting; raw rollout data and an anonymous executable experimental package are separate artifacts, not claimed publicly released here.

ETHICS STATEMENT

Experiments are simulated. Neither terminal success nor a drop proxy certifies deployment safety. Privileged diagnostics are distinguished from RGB-only decisions. Physical recovery consumes the action budget rather than providing cost-free simulator resets. Real-robot use requires independent safety review.

AI USE STATEMENT

Generative AI coding assistance was used for literature synthesis, research planning, implementation and debugging, analysis, result organization, and manuscript drafting. Quantitative results derive from saved experiment artifacts, not invented measurements. Automated checks do not replace scientific review. This working manuscript’s claims, citations, and disclosure remain subject to final human-author verification before submission.

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A LATENT REPRESENTATION AND METRIC DEFINITIONS

A.1 DECODING

Normalized low-dimensional targets are repeated in designated latent frames. The decoder configuration has latent shape $[1, 16, 9, 28, 28]$ (batch, channels, time, height, width). One action frame has 12544 scalars, or 112 complete copies of a 16×7 chunk. Channel-height-width flattening, not spatial action-time tokenization, defines the repeated vectors. For $B^A \in \mathbb{R}^{R_A \times T \times D}$:

$$\tilde{A}_{h,d} = \frac{1}{R_A} \sum_r B_{r,h,d}^A, \quad A = \text{denormalize}(\tilde{A}). \quad (7)$$

Proprio uses repeated 9 D vectors, value its repeated scalar frame, and future RGB the VAE decoder. Trailing incomplete proprio vectors are excluded. Copies within one generation are correlated, not independent model samples.

A.2 SAMPLE AND COPY AXES

Standard deviations use NumPy population normalization, $\text{ddof} = 0$:

$$U_{A,\text{mean}} = \frac{1}{TD} \sum_{h,d} \sigma_{h,d}, \quad U_{A,\text{first}} = \sqrt{\sum_{d=1}^6 \sigma_{1,d}^2}, \quad (8)$$

$$\bar{V} = \frac{1}{K} \sum_i V_i, \quad U_{V,\text{range}} = \max_i V_i - \min_i V_i, \quad (9)$$

$$U_p = \frac{1}{9} \sum_d \text{Std}_i(\hat{p}_{i,d}), \quad U_I = \frac{1}{3HW} \sum_{u,v,c} \text{Std}_i(\hat{I}_i[u,v,c]). \quad (10)$$

NaN-aware value metrics use finite subsets. Future-proprio dispersion uses normalized latent-derived values; image dispersion depends on pixel scale. Internal metrics for a single sample are

$$U_{A,\text{copy}}^i = \frac{1}{TD} \sum_{h,d} \text{Std}_r B_{i,r,h,d}^A, \quad (11)$$

$$U_{p,\text{copy}}^i = \frac{1}{9} \sum_d \text{Std}_r B_{i,r,d}^p, \quad (12)$$

$$U_{V,\text{element}}^i = \text{Std}_\ell(\text{flat}(L_i^V)_\ell). \quad (13)$$

Latent first-action copy disagreement includes all 7 coordinates, unlike the 6D across-sample statistic. A mean-over-samples suffix averages over i ; episode std then aggregates across queries. These axes cannot be conflated.

Joint V_i vary with different actions/futures. Conditional variance $\text{Var}(V \mid x, A_i)$ requires repeats holding action fixed. A common penalty cannot change selection:

$$\arg \max_i [V_i - \lambda U(x)] = \arg \max_i V_i. \quad (14)$$

The historical risk-overconfidence-value predictor averages available training-standardized components

$$\{-z(U_{V,\text{std}}), -z(U_{V,\text{range}}), -z(U_{V,\text{element}}), -0.5z(\bar{V})\}. \quad (15)$$

Its name is not a theoretical definition: these signs favor higher risk for lower dispersion and lower mean value. The denominator is the number of available components. It is not a validated universal risk formula.

A.3 PREDICTION ERROR AND LEAKAGE

For a completely executed prediction horizon:

$$E_p = \|\hat{p}_{t+T} - p_{t+T}\|_2, \quad E_I = \frac{1}{3HW} \|\hat{I}_{t+T} - I_{t+T}\|_F^2. \quad (16)$$

SSIM/LPIPS are optional, not guaranteed present in every campaign. Proprio scales must match. A forecast for $t + 16$ cannot be scored against $t + 8$. The last H8 campaign leaves these fields empty, as with early-terminal H16 chunks; missing is not zero. Post-execution errors inform the next query, not an accident already occurring inside the executed chunk. Whole-episode aggregates likewise use future information. At H16, query 3 starts at 48 and ends at 64; an error at 64 cannot anticipate a drop at 48–56. Variable H/recovery require actual saved timestamps.

Aligned overlap for generate 16/execute 8 is

$$U_{\text{overlap}}(q) = \sqrt{\frac{1}{8D} \sum_{h=1}^8 \sum_d (A_q[h, d] - A_{q-1}[h+8, d])^2}. \quad (17)$$

Disagreement may be beneficial correction; consistency can accompany failure. Endpoint images from consecutive queries usually target different times. Failure is the positive classification label: $\text{TPR} = \text{TP}/(\text{TP} + \text{FN})$, $\text{TN} = \text{TN}/(\text{TN} + \text{FP})$; balanced accuracy is their mean. Feature/threshold selection and normalization belong inside training folds. Holding out seeds but not task/init does not establish new-scene transfer. Event detection requires pre-event lead time.

B CONTROLLER PARAMETERS AND DECODER LAYOUT

For clipped translation u_h , rotation ω_h , and gripper sign b_h :

$$\bar{p}_h = \sum_{\tau \leq h} c_p u_\tau, \quad \bar{R}_h = \exp([c_r \omega_h]_\times) \bar{R}_{h-1}, \quad \bar{R}_0 = I, \quad (18)$$

$$\vartheta(R, S) = \arccos \left(\text{clip} \left(\frac{\text{tr}(RS^\top) - 1}{2}, -1, 1 \right) \right). \quad (19)$$

The broad OSC distance is

$$D(i, j) = \frac{\sum_{h=1}^{16} \gamma^{h-1} [\|\bar{p}_h^i - \bar{p}_h^j\|_2 / c_p + \lambda_r \vartheta(\bar{R}_h^i, \bar{R}_h^j) / c_r + \lambda_g \mathbf{1}(b_h^i \neq b_h^j)]}{\sum_{h=1}^{16} \gamma^{h-1}}. \quad (20)$$

Frozen parameters: $c_p = .05$, $c_r = .5$, $\lambda_r = .5$, $\lambda_g = .25$, $\gamma = .95$, zero alignment window. Command proxies are not true contact-aware poses. Averaging distances over K including zero diagonal only rescales the medoid cost. OSC ties use larger value then smaller index, so selection is not strictly value-free in ties. Raw medoid uses flattened native actions; KeyStone-style adds guarded clustering; KDPE uses induced endpoints.

The decoder campaign’s action control instead uses first 5 actions, discount.9, translation norm/ $\sqrt{3}$, rotation-angle/ π with weight.5, and gripper-sign mismatch with weight.25. It is not the broad OSC control. Native axis-angle commands are not continuous 6 D rotations.

Decoder capture uses final conditional forward, action index 4 of 9, spatial patch 2 and temporal patch 1:196 vectors of dimension 2048. Incidence follows the actual channel-height-width repeated action layout. Cosine uses a 10^{-12} norm floor and clipped dot products. No policy-weight updates or averaging of incompatible actions are performed.

Candidate groups are (1, 999, 998), (2, 997, 996), (3, 995, 994), separate from env seed, and reused at every query. Object/Environment cover 10 tasks, init 0, 1; Position uses x 0.2/y 0.2, init 0. Sixty base configurations times 3 groups yield 180. This is controlled fixed-seed sampling, not new iid noise per query. Modal selection is 652/772(group 0/index 1), 656/788(group 1/index 0), 634/781(group 2/index 2). Equal-episode rates 85.90%, 84.99%, 82.70% show that affinity is not only length weighting. It is visible atq 0. Full/prefix shadow choices differ 7.22%, full/uniform 16.66%; shadow decisions are not separate deployed SRs.

C HORIZON AND COMPUTE CONTROLS

H16 controls below use exactly the 120 H8 configurations. Only 360 new H8 rollouts are added; reused controls are not new independent data. Full-minus-max H8: -1.67 pp, CI $[-6.67, 2.78]$, 9

rescues/10 harms. Prefix-minus-max H8 has zero macro effect despite 9 rescues/7 harms because factor weights differ from pooled episode weights. Prefix-minus-full H8: +1.67 pp, CI $[-.56, 3.89]$. MaxH 8-minus-maxH 16: -5.56 pp, CI $[-13.89, 1.67]$. FullH 8-minus-decoderH 16: -8.33 pp, CI $[-19.44, 1.11]$. Mean max-value candidate calls H16/H8 are 29.0/55.9 Object, 40.2/83.1 Environ-

Table 5: Matched horizon results. Factor denominators 30/30/60; total successes out of 120. Macro is not micro.

Controller	Obj.	Env.	Pos.	Total	Macro (%)
Max-value H16	28	15	31	74	65.00
Decoder H16	30	15	29	74	66.11
Max-value H8	28	13	25	66	59.44
Full decoder H8	27	12	26	65	57.78
Prefix decoder H8	27	12	29	68	59.44

ment, 46.75/94.3 Position. Episode length affects these counts. Shared hardware, instrumentation, and cache differences prevent interpreting wall time as a controlled production-latency comparison.

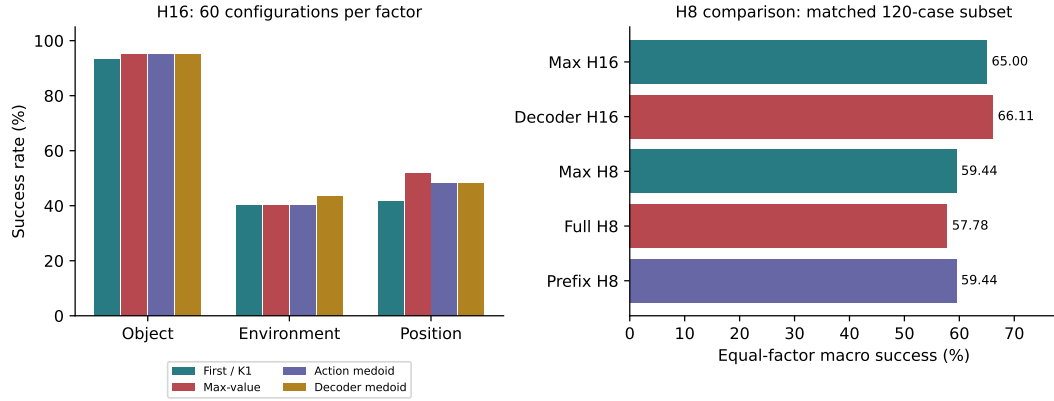


Figure 2: H16 factor rates (left) and matched horizon subset (right). Bars are point estimates; paired intervals are in the text. Populations differ between panels.

D CANDIDATE REPEATS AND RECOVERY DETAILS

D.1 SAVED CANDIDATES AND FEEDBACK

The 36 snapshots are at $q/3$ ($t = 0, 48$) from 18 case/task/init configurations sharing 10 underlying task/init clusters. Eight candidates give 288 strictly replayed branches with a common type of K1/H16 continuation. Nested first 4 yields max 15/36 and oracle 19/36, not a deployed K4 benchmark. The 1080-branch repeat series contains 864 candidate branches (36 states, 8 actions, 3 suffixes) and 216 feedback branches (3 arms, 2 suffixes per state). Candidate outcomes depend on continuation, budget, and noise, not just the first chunk.

The 690-branch follow-up includes two local pools and 640 branches on eight neighbors (tasks 0/2, init 9-12, Positionx 0.2, 8 actions, 10 suffixes). Pool 13 alternatives give 10/10 versus 1/10 max. Pool 11 gives 6/10 versus 4/10, 4 rescues/2 harms, $p = .6875$, not confirmed. Two neighbors remain all-fail over 80 branches each, not proof of impossibility. Matched- K feedback uses 24 states/12 clusters, 5 arms:120 branches. FreshK 4 changes 17/24 actions, continuity only 4/24; both succeed 13/24, versus 15/24 open 16.

D.2 RECOVERY DEVELOPMENT

Privileged regrasp succeeds 62/80 versus 7/80 H4 and 8/80 blind lift, a conditional oracle result, not full benchmark SR. RGB development restores 43/72 cases across 47 task/init groups. P3c uses init 35–39 on 8 selected cells, frozen localizer/gate, full H16 queries 0–3 then 8 actions of q 4, branching at 72. The 13/40 versus 24/40 result has 12 rescues/1 harm, $p = .00342$. Same-cell 6/40 versus 25/40 has CI [30, 65] pp; new-cells 26/35 versus 31/35 has CI [−2.9, 31.4] pp. New recovery-test cells were not necessarily outside localizer training. Full versus retreat-only gives 56/75 versus 36/75.

A historical nonphysical fallback was found but did not activate in audited main P3c/P3d branches. New protocols do not undo physical steps. Some old illustrative videos are seeded replays rather than original records. On 48 states, continue/full/probe/verified counts are 26/35/26/30; 7 of 9 held decisions are false-held because crops follow the gripper. Grounded-mask screening on 48 newstates gives full 33, conservative/probe-always 35, identical outcomes and 47 identical trajectories. Ten false-held errors vanish on 27 common probes, but 8 become unknown; a separate verifier control gain is not isolated.

Another 48-state delayed test gives 33 versus 36 full successes; the fresh gate cancels 14 of 21 eligible recoveries, confounding timing. A 96-prefix/48-init-cluster timing test (480 main+10 smoke) gives continue 63, immediate/diagnostic 76, fresh/checked 74. Checked-minus-immediate: −2.08 pp, CI [−7.29, 3.13], 3 rescues/5 harms, Holm 1. All 17 shared failures lie outside the old trigger.

D.3 LATEST OBSERVATION CONTRACT

Tasks 2/5/9, x/y 0.1/0.2 define 12 selected Position cells. Init 46–49, two prefix realizations and two suffixes give 192 branches/arm, 48 initclusters; prefixes were collected previously. Ninety-eight cases pass old eligibility; internal checks permit 70 full recoveries. Ninety-two receive refresh probes; 20 extra regrasp maneuvers are added by the new branch. Three-step probe target 8 cm produces about 3.2 cm actual motion. Preserve keeps clipped previous gripper command; open uses −1. All refresh arms retain old-gate recovery.

No six-way predeclared pooled contrast passes Holm.05. Primary preserve+regrasp-minus-physical CI [−6.25, 0] pp, Holm 1. Preserve+regrasp-minus-preserve-only CI [−11.46, −5.21] pp, Holm.18948. Post-hoc preserve-only-minus-physical: +5.208 pp, CI [3.125, 7.8125], 11 rescues/1 harm, cluster-sign $p = .09587$. Net ten successes arise in task 9/x 0.2(1/16 to 11/16); task 2/y 0.2 has one rescue and one harm; other net effects are zero. GT physical/calibrated workspace arms are privileged diagnostics.

All 16 extra-regrasp harms have identical common-probe actions/inputs, eight prefixes and six initclusters. Contact persists for 3 probe steps, object and gripper moving together. Opening at 76 precedes drops at 82–85, before maneuver end 95–97. These paired trajectories support localized intervention-induced failure, not an online GT-based detector.

E STATISTICS, HISTORICAL EVIDENCE, AND INTEGRITY

Valid 199 uses factor-balanced effects and task/init-cluster bootstrap. Decoder resamples tasks within factors, retaining seed groups; intervals are unadjusted. Q 4/P3c cluster by init. Observation intervals use stratified fixed-cell init bootstrap, while significance uses cluster-sign tests and Holm over six contrasts. Narrow bootstrap intervals do not override corrected-test decisions or establish new-cell transfer. Episode McNemar is descriptive under seed dependence and not a separate test of macro-SR.

Table 6 retains predecessor results with distinct populations; they are not additional independent main-test observations.

Cosmos’s standard LIBERO policy score and separate planning result are not our joint-value PRO protocol (Kim et al., 2026). KeyStone’s sampling budget and KDPE’s action endpoints differ from our adaptations (Dai et al., 2026; Rosasco et al., 2025). Flow-VLA UQ concerns data acquisition/fine-tuning (Römer et al., 2026); Sentinel concerns monitoring and Rewind-IL adds recovery (Agia et al., 2025; Zheng et al., 2026). Negative transfer here does not refute those results.

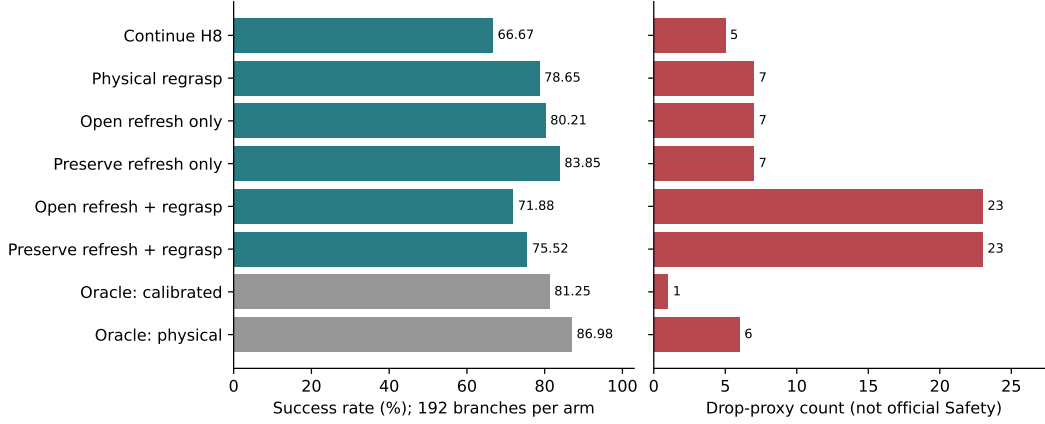


Figure 3: Observation-contract point estimates and local drop counts. Gray denotes privileged oracles. The best deployable point estimate is post-hoc and one-cell concentrated; the primary method is not confirmed.

Table 6: Historical evidence. Counts are control versus method unless labeled. These rows are not a shared benchmark or a formal meta-analysis.

Study	Observation	Scope
Early 12-seed case	Max 6, action 6, value 7, combined 4	Exploratory selected case
Static penalties, 10 steps	52/100 vs 56/100	CI $[-5, 13]$ pp
Adaptive horizon, 6 cases	115/180 vs 143/180	Replay disagreement 4.4–9.4%
Selection/horizon factorial	100/168 vs 121/168	Init-CI $[4.8, 20.8]$; case-CI $[-4, 29]$ pp
Temporal overlap	312 rollouts, AUROC near.50	Some event labels invalidated
Boundary screen	264 rollouts, best AUROC.613	Interval includes.5
Frozen ranker	Offline regret -43.1%	Closed-loop 165/360 vs 164/360
Surrogate requery	146/240 vs 147/240	Older requery 161/240
AR pilot	10 states, 80 candidates, 30%SR	Not original Cosmos planning
Signed feedback transfer	240 pairs, AUROC.398	No transferable benefit predictor
Invariant CATE reserve	400 pairs, 57%vs 55.75%	No gain over commit
Residual dynamics	5668 transitions	Quadratic-JRD collapses after clamp
Official LIBERO-Safety	144 rollouts, 0 safe-success	Four violations; missing semantic assets

The decoder audit validates 1440 videos/297109 frames, 23933 queries, selected actions, fresh inputs, horizons, and 180 exact initial groups. Observation auditing decodes 1560 videos/183325 frames including 24 smoke videos; 1536 main outcomes and 184 common-probe pairs are checked. Together 3000 videos/480434 frames include technical and dependent branches. Earlier results rely on recorded reports/audits, not a claim to re-audit every old raw trajectory. Generated tables retain source hashes.

F PROSPECTIVE TESTS, NOT ADDITIONAL RESULTS

Freeze contact-preserving refresh/recovery before new task/init/cell tests. Compare continuation, old recovery, and matched probes; report SR, drops, false interventions, and compute. A possible objective is

$$b^*(x) = \arg \max_b [\widehat{\Pr}(Y = 1 \mid x, b) - \lambda \widehat{\Pr}(\text{harm} \mid x, b) - \eta \text{cost}(b)], \quad (21)$$

where b is continue, refresh, or recover. This is a direction, not an evaluated new algorithm. It requires paired training branches, cluster-separated calibration, and a frozen holdout. Decoder refinements should first pass new seed-bank/permutation/fixed-index controls. When pools lack useful actions, proposal expansion or grounding should precede more reranking hyperparameter searches.